

ST ALBERT, AB, Canada

WMO: 711215

Lat: 53.6920N

Lon: 113.6196W

Elev: 2254

StdP: 13.54

Time Zone: -7.00 (NAM)

Period: 10-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-23.8	-17.6	-29.9	1.1	-23.7	-23.5	1.6	-17.2	24.2	24.4	20.7	23.8	4.0	110	0.668

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.2	81.7	62.8	79.0	61.8	76.3	61.0	66.9	77.2	64.8	75.0	62.9	73.0	7.9	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.9	93.2	73.6	60.7	86.1	70.3	58.8	80.4	68.3	32.8	76.7	31.1	75.0	29.6	72.9	76.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.4	19.1	16.8	DB	-32.8	88.0	5.9	2.8	-37.1	90.0	-40.5	91.6	-43.8	93.2	-48.1	95.2	(4)
(5)			WB	-33.0	70.9	5.9	2.7	-37.2	72.9	-40.7	74.4	-44.0	75.9	-48.2	77.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	37.2	14.0	11.6	22.8	37.8	52.5	58.6	62.1	60.1	51.3	38.8	22.0	12.4
	DBStd	21.70	15.02	15.84	14.37	9.62	7.96	5.27	4.70	5.29	7.99	8.12	12.86	13.27
	HDD50	5922	1115	1075	843	376	66	5	0	2	79	355	841	1165
	HDD65	10219	1580	1495	1308	815	391	203	114	165	413	814	1291	1630
	CDD50	1230	0	0	0	10	143	261	376	316	117	7	0	0
	CDD65	54	0	0	0	0	4	10	25	14	1	0	0	0
	CDH74	1107	0	0	0	4	156	162	386	283	114	2	0	0
	CDH80	161	0	0	0	0	24	17	63	36	21	0	0	0
(14) Wind	WSAvg	7.7	7.8	7.5	7.4	8.8	9.0	8.5	7.0	6.4	7.2	8.2	7.0	7.2
(15) Precipitation	PrecAvg	16.9	0.7	0.4	0.6	1.1	2.0	3.0	3.4	2.2	1.3	0.8	0.7	0.6
	PrecMax	23.5	2.8	1.1	1.3	1.9	4.0	6.1	5.6	4.8	3.2	2.2	2.8	1.0
	PrecMin	9.5	0.1	0.0	0.2	0.1	0.4	0.6	0.8	0.9	0.2	0.2	0.2	0.2
	PrecStd	3.2	0.6	0.2	0.3	0.5	0.9	1.4	1.2	1.0	0.9	0.5	0.6	0.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.4	44.3	53.6	72.8	83.5	82.3	85.8	83.9	83.4	68.2	53.2	42.1
		MCWB	37.0	38.2	42.1	51.6	56.6	61.0	68.0	63.4	60.8	52.5	43.6	36.1
	2%	DB	40.9	39.5	47.8	64.3	78.8	78.7	81.4	79.8	77.0	61.5	46.0	38.1
		MCWB	35.8	34.4	39.7	47.0	54.7	60.7	66.0	64.2	58.5	47.5	39.1	33.4
	5%	DB	38.1	36.6	43.5	58.8	74.0	75.0	78.3	77.3	72.4	57.1	41.7	34.7
		MCWB	34.0	32.6	37.2	44.4	53.2	58.8	64.3	63.4	57.2	46.1	37.0	30.7
	10%	DB	34.9	33.5	39.7	53.9	69.7	71.8	75.3	74.3	67.2	52.6	37.8	30.6
		MCWB	31.7	30.5	34.7	42.2	51.2	57.5	62.8	61.8	54.6	43.5	34.5	27.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.5	38.4	44.0	51.7	58.2	65.6	70.5	69.1	62.9	52.7	44.9	36.7
		MCDB	42.5	44.2	51.2	71.5	76.0	76.8	82.3	78.6	78.0	67.1	51.4	41.4
	2%	WB	36.0	35.1	40.0	47.6	56.2	62.6	67.7	66.5	60.1	48.8	40.4	33.5
		MCDB	40.3	38.7	47.1	62.9	73.5	74.5	78.3	76.6	74.6	59.4	45.3	37.5
	5%	WB	34.2	32.9	37.4	45.0	54.4	60.6	65.6	64.3	57.5	46.5	37.2	30.9
		MCDB	37.7	36.2	43.0	57.2	71.1	71.2	75.9	74.3	71.1	54.6	40.7	34.8
	10%	WB	31.6	30.5	35.0	42.6	52.5	58.9	63.8	62.5	55.4	44.4	34.7	27.5
		MCDB	35.0	33.6	39.0	52.8	67.7	69.4	73.4	72.5	65.6	51.6	37.9	30.4
(35) Mean Daily Temperature Range	5% DB	MDBR	17.7	19.4	19.2	19.7	24.8	21.5	22.2	24.0	23.9	20.8	15.7	16.6
		MCDBR	16.6	19.6	22.7	29.5	32.1	27.1	28.1	28.8	33.9	28.2	19.2	19.1
	5% WB	MCWBR	13.0	15.2	15.3	15.4	13.9	12.5	15.0	15.1	17.1	16.1	13.4	15.5
		MCDWR	16.7	19.0	21.3	27.4	29.1	23.8	25.8	26.6	31.7	25.6	16.7	19.2
(40) Clear-Sky Solar Irradiance	taub		0.255	0.267	0.310	0.326	0.341	0.351	0.354	0.360	0.307	0.281	0.253	0.238
		taud	2.308	2.352	2.307	2.392	2.395	2.382	2.420	2.352	2.500	2.549	2.456	2.319
	Ebn at Noon		233	269	277	288	287	285	281	272	278	264	237	217
		Edh at Noon	18	24	31	33	35	36	34	34	26	20	16	14
(44) All-Sky Solar Radiation	RadAvg		302	631	1117	1480	1809	1904	1943	1600	1105	633	303	222
	RadStd		33	48	84	137	135	125	98	111	114	75	34	20

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (67 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page